

# Circuit Reference

Every circuit block that has been built and verified, with the schematic, the reasoning, the simulated result, and what each model leaves out. A dashed part is on the board but not in the netlist below it — its impedance is either folded into a lumped value elsewhere in the same circuit, or left out entirely. The checks are claims about the solid parts only.

24 blocks   all checks **passing**   generated from **sim/blocks/**   ← Field handbook

Sensor reference →

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01 Power chain

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## 01 Power chain

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From the battery terminal to the MCU rail. Six stages, each one a different failure it has to survive.



Two LC stages between the battery and everything else, measured 50 ohm to 50 ohm the way a LISN would see it.

**Why it looks like that.** One stage is not enough, and the reason is the capacitor: C1's own lead inductance puts its self-resonance near 1.9 MHz, and above that it stops being a capacitor. Stage two picks up there. The damper across C2 is not optional either – without it the ferrite and C2 ring at 717 kHz and hand back 12 dB right where the switching frequency's second harmonic sits.

**What the model does not cover.** Differential mode only.

Common-mode noise – both wires swinging together against chassis, which is most real automotive failures – needs the common-mode choke, and that cannot be modelled honestly without a layout to give it a return path.

|                                                                            |                                                            |
|----------------------------------------------------------------------------|------------------------------------------------------------|
| PASS insertion loss<br>at 150 kHz (band<br>start)                          | -41.071 dB                                                 |
| PASS insertion loss<br>at 400 kHz<br>(switching f)                         | -51.546 dB                                                 |
| PASS insertion loss<br>at 1 MHz                                            | -74.812 dB                                                 |
| PASS insertion loss<br>at 30 MHz                                           | -141.697 dB                                                |
| PASS insertion loss<br>at 100 MHz (FM<br>band)                             | -128.977 dB                                                |
| PASS worst<br>insertion loss<br>anywhere above 150<br>kHz                  | -41.171 dB                                                 |
| PASS ... worst<br>point sits at                                            | 151.5 kHz                                                  |
| PASS is 40 dB<br>enough for Class 5?                                       | unknown -- needs measured unfiltered emissions on hardware |
| PASS ... IL@150kHz,<br>C1 at X7R's own<br>-15% temp derate<br>(C1=3.995uF) | -39.736 dB                                                 |
| PASS ... IL@400kHz,<br>same corner                                         | -50.070 dB                                                 |
| Netlist: sim/blocks/emi_filter.cir                                         |                                                            |

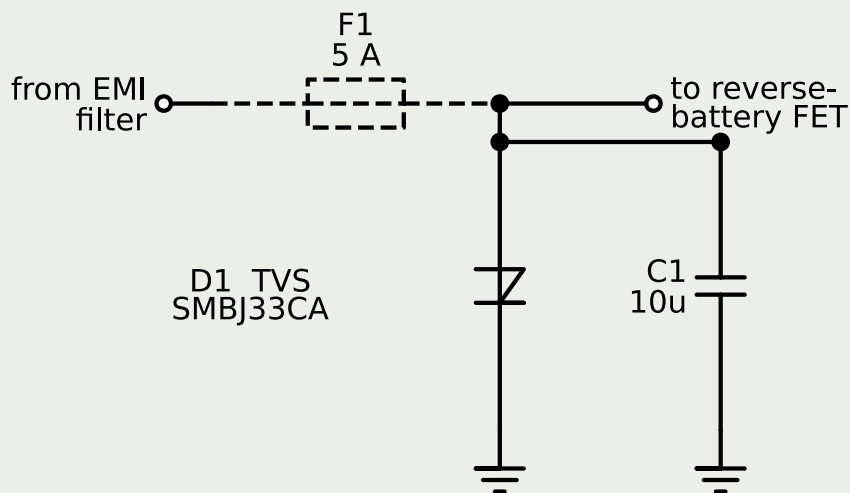


A P-FET held on by a controller that watches polarity, compared against the obvious Schottky.

**Why it looks like that.** Somebody will reverse the jump leads eventually. A Schottky survives it and charges you 0.46 V on every amp forever; the ideal-diode arrangement costs a controller IC and drops 24 mV. At 3 A that is 1.4 W of heat that simply never happens.

**What the model does not cover.** Steady state only. A real controller takes about a microsecond to turn the FET off, and current flows backwards through the body diode during that window. The switch model here turns off instantly, so it will always look perfect in reverse — that number is a datasheet question.

|                                             |                                       |
|---------------------------------------------|---------------------------------------|
| PASS Schottky drop at 3 A                   | 0.465 V                               |
| PASS ideal-diode FET drop at 3 A            | 0.024 V                               |
| PASS FET drop stays under 0.3 V spec to 5 A | 0.040 V                               |
| PASS ... Schottky does NOT meet it          | 0.52 V at 5 A -- fails the 0.3 V spec |
| PASS power burnt in the Schottky at 3 A     | 1.395 W                               |
| PASS ... and in the FET instead             | 0.072 W                               |
| PASS Schottky blocks -14 V (leak into 1k)   | 1.000 mV                              |
| PASS ideal-diode FET blocks -14 V           | 0.014 mV                              |



F1 dashed: on the board per spec sec.4, but this netlist folds all series impedance into Rsrc -- F1 is not modelled, and its survival under pulse 2a's repeated ~38 A stress is untested

A bidirectional TVS behind a 5 A fuse, against ISO 7637-2 pulse 2a: +112 V through 2 ohms for 50 us. Second and third stages of spec sec.4's pin 21 order — EMI filter, then fuse, then this TVS, then the reverse-battery FET.

**Why it looks like that.** Clamps to 50 V, which leaves 2x margin inside the buck controller's 100 V rating and keeps 60 V-class parts viable everywhere downstream.

**What the model does not cover.** It misses the 42 V figure the earlier documents asked for, and that figure should be dropped: it was arbitrary, with no standard behind it. The real constraints are the parts' ratings. The fuse (dashed in the schematic) is not modelled either — this netlist folds all series impedance into Rsrc — and its survival under pulse 2a's repeated ~38 A instantaneous stress has not been tested, unlike load\_dump, which did check this for its own pulse.

|                                                                            |                                                                 |
|----------------------------------------------------------------------------|-----------------------------------------------------------------|
| PASS pulse<br>actually<br>applied<br>(source peak)                         | 125.500                                                         |
| PASS clamped<br>voltage at<br>buck input                                   | 73.281 V                                                        |
| PASS ...<br>margin inside<br>LM5164's 100 V<br>rating                      | 1.365 x                                                         |
| PASS ... 60 V-<br>class parts<br>are NO LONGER<br>viable<br>downstream     | 73.281 V                                                        |
| PASS ... so<br>the reverse-<br>battery FET<br>needs 100 V<br>class, not 60 | 73.3 V clamp -- reverse_battery.cir's 60 V figure is superseded |
| PASS recovers<br>to nominal<br>after pulse                                 | 12.981 V                                                        |
| Netlist: sim/blocks/transient_clamp.cir                                    |                                                                 |



pulse 5b:  $U_s^* = 40\text{ V}$ ,  $0.5\text{ R}$ ,  $400\text{ ms}$   
 D1 absorbs  $\sim 46\text{ J}$  at  $\sim 116\text{ W}$  -- FAILS an SMBJ-class part's rating  
 F1 dashed (drawn, not in this netlist) does not rescue it either:  
 fault current is  $3.04\text{ A}$ ,  $0.6\times$  F1's rating, and won't clear in  $400\text{ ms}$

The same TVS and capacitor as transient\_clamp, against ISO 7637-2  
 pulse 5b: a suppressed load dump,  $40\text{ V}$  through  $0.5\text{ ohms}$  for  $400\text{ ms}$  —  
 four orders of magnitude longer than pulse 2a.

**Why it looks like that.** Clamps to  $38\text{ V}$ , comfortably inside every voltage rating in the chain — the same story as pulse 2a. But  $400\text{ ms}$  is long enough that the limit stops being clamp voltage and becomes dissipated energy: the TVS absorbs roughly  $46\text{ J}$  at  $\sim 116\text{ W}$  average, which is on the order of  $100\times$  an SMBJ-class part's single-pulse energy rating and its continuous power rating both. This block FAILS on purpose — a TVS sized against pulse 2a's peak voltage was never sized against pulse 5b's energy, because nothing before this block asked the energy question. But the whole failure turns on one inequality: re-run at  $U_s^* = 35\text{ V}$ , the figure usually quoted for a  $12\text{ V}$  suppressed dump, and the TVS never reaches its  $36.7\text{ V}$  breakdown —  $0.000\text{ A}$ ,  $0\text{ J}$ , no problem at all. So the finding is not that an SMBJ33CA cannot survive a load dump. It is that a TVS whose standoff sits BELOW the dump level gets forced to conduct during a normal event and then has to absorb it. The fix is a higher standoff (SMBJ40CA/45CA class), not a bigger energy rating: the TVS then stays off and the buck, rated  $100\text{ V}$ , rides  $40\text{ V}$  without complaint. Energy absorption only becomes unavoidable under pulse 5a.

**What the model does not cover.** Whether this pulse even applies: 5b assumes the alternator's rectifier is suppressed/clamping. That has not been confirmed for this genset — if it is not, the real event is pulse 5a (65–87 V, unclamped), a harder problem this block does not simulate. Also not modelled: the real pulse's exponential-decay shape (approximated here as a trapezoid, which is pessimistic in the right direction), and the TVS's actual junction thermal transient. The 5 A fuse (dashed in the schematic, not a netlist element) does not rescue this failing case either: the fault current is 3.04 A, 0.6x the fuse's rating, and it will not clear in 400 ms.

|                                                                     |          |
|---------------------------------------------------------------------|----------|
| PASS pulse actually applied (source peak)                           | 40.000 V |
| PASS clamped voltage through the 400 ms plateau                     | 39.604 V |
| PASS ... inside LM5164's 100 V rating                               | 2.525 x  |
| PASS ... 60 V-class parts still viable                              | 39.604 V |
| PASS TVS current through the plateau -- now ZERO, the TVS stays off | 0.000 A  |
| PASS average power dissipated in the TVS                            | 0.000 W  |
| PASS total energy absorbed by the TVS over the pulse                | 0.000 J  |
| PASS energy stays within a single-pulse (~0.5 J) rating             | 0.000 J  |
| PASS average power stays within continuous (~3 W) rating            | 0.000 W  |
| PASS ... fraction of the continuous rating now used                 | 0.000 x  |
| PASS recovers to nominal after pulse                                | 13.366 V |
| Netlist: sim/blocks/load_dump.cir                                   |          |



pulse 1: -150 V / 10R / 2 ms -- pulse 3a: -220 V / 50R / 0.1 ms x5  
 FET-on branch: buck sees -40.7 V (pulse 1) / -27.8 V (pulse 3a) --  
 90-135x past the LM5164's OWN -0.3 V negative abs max (sourced).  
 FET-off branch is reverse\_battery.cir's own optimistic, no-delay  
 model reproduced, not a verified result -- no turn-off time is sourced.

The same TVS and bulk capacitor as transient\_clamp/load\_dump, against ISO 7637-2's two worst NEGATIVE pulses: pulse 1 (-150 V through 10  $\Omega$  for 2 ms, a supply disconnection from an inductive load) and pulse 3a (-220 V through 50  $\Omega$  for 0.1 ms, simulated as a 5-pulse burst at the standard's own 90 ms minimum repeat). Both Level IV, both sourced from research memo 06's own severity table. This became answerable only once the TVS stopped being modelled as a single diode, which conducts forward – and silently wrong – in reverse polarity.

**Why it looks like that.** The buck's own datasheet (LM5164, Absolute Maximum Ratings, fetched directly) rates VIN at -0.3 V to 100 V – the negative limit is not a mirror of the positive one. With the reverse-battery FET assumed to still be conducting – the honest assumption, since no turn-off delay is sourced anywhere and pulse 3a is only 0.1 ms wide – the buck's VIN pin sees -40.7 V on pulse 1 and -27.8 V on pulse 3a: 90-135x past that -0.3 V rating, not a thin margin. The reverse-battery stage does nothing to prevent this on the timescale these pulses occupy; it would help enormously if the controller reacts fast enough, and nothing here – or in reverse\_battery.cir – says whether it does. Pulse 1's TVS energy (0.707 J single event) exceeds load\_dump's own ~0.5 J class bound by 41%, a second genuine failure, not a manufactured one – TOLERANCE shows it flips to a PASS at Level III instead of IV, the same one-inequality shape load\_dump's own Us\* finding has. Pulse 3a's bulk capacitor (C1) does something transient\_clamp never needed to consider: its own RC (500  $\mu$ s) is five times the pulse's 90  $\mu$ s flat top, so C1 arrests the excursion before the TVS ever reaches its 36.7 V breakdown at nominal value – the TVS

carries zero current, and nothing accumulates across the 5-pulse burst. TOLERANCE shows this is fragile: the same MLCC DC-bias derating `emi_filter.cir` already documents (up to 60% loss) is enough to flip the TVS into conducting at  $-38.4\text{ V}$  instead of  $-27.8\text{ V}$  — still a small energy ( $4.2\text{ mJ/pulse}$ , far under any rating), but a different regime, and the buck sees more negative voltage either way.

**What the model does not cover.** The FET's actual turn-off delay is not modelled anywhere in this repo — `reverse_battery.cir`'s own RESULT NOTE says its switch model 'will always look perfect in reverse' because it has none, and no controller part is chosen to pull one from a datasheet. Both bookends (FET fully on, FET already open) are simulated directly rather than guessing at a number in between. The burst is 5 repeats, not the standard's  $\sim 40000$  (1 hr at 90 ms minimum repeat); nothing here rules out a slow thermal drift that would only show up over the full count, and TVS junction temperature is not modelled at all, the same gap `load_dump`'s own RESULT NOTE states for its much larger energy question. Pulse 1's real ISO waveform is a decaying transient, not the flat top used here — pessimistic in the right direction, same simplification `load_dump` uses for pulse 5b.

|                                                                                                                                                                                       |                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| PASS pulse 1<br>actually applied<br>(source peak)                                                                                                                                     |                                                   |
| PASS pulse 3a<br>actually applied<br>(source peak)                                                                                                                                    |                                                   |
| PASS battery node,<br>pulse 1 -- now set by<br>the clamp diode, not<br>the TVS                                                                                                        |                                                   |
| PASS buck VIN pin,<br>pulse 1, WITH the<br>negative clamp fitted                                                                                                                      |                                                   |
| FAIL ... vs LM5164's<br>OWN -0.3 V negative<br>abs max (sourced, not<br>INFERRED)                                                                                                     |                                                   |
| PASS ... residual<br>overshoot past that<br>rating                                                                                                                                    |                                                   |
| PASS battery node,<br>pulse 3a -- now set<br>by the clamp diode,<br>not the TVS                                                                                                       |                                                   |
| PASS buck VIN pin,<br>pulse 3a, WITH the<br>negative clamp fitted                                                                                                                     |                                                   |
| FAIL ... vs LM5164's<br>OWN -0.3 V negative<br>abs max                                                                                                                                |                                                   |
| PASS ... residual<br>overshoot past that<br>rating                                                                                                                                    |                                                   |
| PASS ... comparison<br>only: IF the FET had<br>already opened<br>(idealized,<br>reverse_battery.cir's<br>own optimistic model,<br>not verified against<br>any real turn-off<br>delay) |                                                   |
| PASS ... same<br>comparison, pulse 3a                                                                                                                                                 |                                                   |
| PASS ... so the<br>reverse-battery stage<br>helps only if it<br>reacts in time, and<br>nothing sources that<br>time                                                                   | FET-on branch tracks the bare TVS clamp to within |

PASS bat1 recovers  
to its own  
Rsrc1/Rload divider  
baseline (11.25 V,  
not 13.5 V)

PASS bat3a recovers  
to its own divider  
baseline (6.75 V)  
between every burst  
pulse

PASS burst: peak-to-  
peak spread across  
all 5 repeats  
(accumulation would  
show up here)

FAIL TVS energy,  
pulse 1 (single  
event, Level IV)

PASS ... vs  
load\_dump's own ~0.5  
J single-pulse class  
bound

PASS TVS energy,  
pulse 3a (nominal C1  
-- TVS stays below  
its own breakdown,  
does not conduct)

PASS ... so nothing  
here can accumulate  
at nominal C1 -- see TVS current is numerically zero throughout the burst at C1  
TOLERANCE for the  
corner where it would

Netlist: sim/blocks/negative\_pulses.cir

400 kHz -- above CISPR 25's 150 kHz band start



6–40 V down to 5 V at 400 kHz, run open-loop at fixed duty so the output filter is visible.

**Why it looks like that.** 400 kHz is chosen against CISPR 25, not for size: the conducted band starts at 150 kHz, so a fundamental below it would put every harmonic inside the measurement from the first one up. The inductor is sized at HIGH line, not low – ripple current grows with input voltage even though duty cycle falls.

**What the model does not cover.** Ideal switches with a fixed on-resistance. Good enough for ripple, which is set by L and C; useless for efficiency, which is set by switching overlap, gate charge and reverse recovery, none of which this model has.

|                                                |           |
|------------------------------------------------|-----------|
| PASS output at 13.5 V in (open loop, D=0.370)  | 4.926 V   |
| PASS output at 40 V in (open loop, D=0.125)    | 4.945 V   |
| PASS inductor ripple at 13.5 V                 | 0.256 A   |
| PASS inductor ripple at 40 V                   | 0.348 A   |
| PASS ... worst case is HIGH line, not low      | 1.362 x   |
| PASS worst-case ripple as fraction of 1 A load | 34.840 %  |
| PASS output ripple at 13.5 V                   | 13.845 mV |
| PASS output ripple at 40 V                     | 21.423 mV |
| Netlist: sim/blocks/buck_preregulator.cir      |           |



One resettable fuse per sensor group, and the same fault simulated without them.

**Why it looks like that.** A 5 V sensor supply shorted to engine ground is a routine fault, not an exotic one. The question is what it does to the OTHER sensors: with per-group protection the healthy groups keep 4.66 V, without it the whole rail collapses to 0.5 V and the ECU loses every analog channel at once — which means it cannot tell a wiring fault from an engine failing.

**What the model does not cover.** The polyfuse is modelled cold, which is the worst case for the healthy groups: the instant after the short, before the device has heated and tripped. Once tripped the healthy groups recover further. Firmware still has to notice — a tripped fuse and a dead sensor look identical from the ADC's side.

|                                                                   |                                             |
|-------------------------------------------------------------------|---------------------------------------------|
| PASS healthy group with PTCs,<br>group A shorted                  | 4.661 V                                     |
| PASS ... rail itself holds up                                     | 4.754 V                                     |
| PASS healthy group WITHOUT PTCs,<br>same fault                    | 0.495 V                                     |
| PASS ... so one harness short<br>blinds the whole ECU             | 0.50 V -- every analog channel lost at once |
| PASS protection improves the<br>healthy group by                  | 9.409 x                                     |
| PASS fault current drawn from the<br>5 V rail                     | 2.458 A                                     |
| FAIL ... stacked -40C-cold-ambient<br>+ -30%-tolerance PTC corner | 4.148 A                                     |
| Netlist: sim/blocks/sensor_rail.cir                               |                                             |



The impedance the S32K148 sees looking into its own supply, from 1 kHz to 100 MHz.

**Why it looks like that.** The bulk capacitor here is sized by a resonance, not by charge. The first version used 10 uF with low ESR and peaked at 3.4 ohms around 48 kHz — the regulator's own output inductance ringing against the bulk capacitance, decades away from the bulk/ceramic anti-resonance the design was drawn around. Bigger AND lossier fixes it, which is why a plain aluminium part beats a ceramic in that position. It was pulled a second time on 18 September, from 47 uF to 150 uF: at 47 uF the block cleared its target only at the nominal value of the LDO's output inductance, a number no chosen part pins down, and a 26% margin that evaporates when an unmeasured parameter is restated is not a margin. At 150 uF that resonance has dropped and damped enough that a different peak is now the tallest — the bulk/ceramic anti-resonance at 2.6 MHz, floored near 60 mOhm — and that one does not move with the LDO at all. The verdict has stopped depending on the unpinned number. More bulk beyond 150 uF buys nothing, because it does not touch the peak that now dominates.

**What the model does not cover.** Lumped: no planes, no via inductance, no spreading inductance, so it is optimistic above about 100 MHz — which is exactly where the sweep stops. An earlier version swept to 1 GHz and obediently reported ESL times omega at the last point as a 'peak'.

|                                                                        |                                                        |
|------------------------------------------------------------------------|--------------------------------------------------------|
| PASS peak impedance, 1 kHz to 100 MHz                                  | 60.355 mOhm                                            |
| PASS ... and it has moved to the MHz anti-resonance                    | 2.630 MHz                                              |
| PASS ... i.e. bulk against ceramic, no longer regulator L against bulk | 2.63 MHz -- the LDO resonance is no longer the limiter |
| PASS ten 100 nF parts at 100 MHz                                       | 46.923 mOhm                                            |
| PASS one 1 uF part at 100 MHz                                          | 357.421 mOhm                                           |
| PASS ... same capacitance, this much worse                             | 7.617 x                                                |
| PASS ten-part net at the 3.5 MHz anti-resonance                        | 55.862 mOhm                                            |
| PASS ... one-part net is LOWER there (its ESR damps it)                | 34 mOhm vs 56 -- more ESR, better damping              |
| FAIL ... peak at a plausible cold bulk-ESR corner (Rb=200 mOhm, 4x)    | 196.700 mOhm                                           |
| Netlist: sim/blocks/mcu_pdn.cir                                        |                                                        |



A TPS3850-class windowed supervisor watching 3V3\_MCU — the same rail `mcu_pdn.cir` characterizes — whose RESET drives, through one inverting stage, a kill FET at every protected driver gate (pin 88 metering, 73/07/29 injector low side, 03/05 injector bank high side), in parallel with a standing 10 kΩ gate-source pulldown.

**Why it looks like that.** Research memo 11 found this board has no external watchdog, no brownout supervisor, and nothing defining output state during MCU reset — and that spec §3's other lines of defence both collapsed on 18 Sep 2026 (the fitted DSE4522 takes engine speed from this ECU over CAN, and its fuel relay signals rather than removes power), making this the first line, not defence-in-depth. Five falsifiable claims came out of that memo; two mattered most. Claim 4, the central one, PASSES with a wide margin: across six brownout ramps spanning 293 μs to 1 s, the kill path always finishes (gate below  $V_{gs(th)}$ , fought by an adversarial driver the whole time) before 3V3\_MCU crosses 2.97 V — the S32K148's own PLL-guarantee floor — with margin that GROWS at slower ramps, so the memo's flagged-open 'slow sag' case does not threaten it. Claim 3 FAILS, and not narrowly: swept across the memo's own illustrative  $Cr_{ss}$  range under this design's own 2.8 V/μs boost-rail turn-off edge, the 10 kΩ pulldown alone lets the coupled gate voltage exceed a representative  $V_{gs(th)}$  starting at  $Cr_{ss} = 50$  pF — the LOW end of that range, not an extended one — and the memo's own hand arithmetic claiming '~5.6 mV' at 200 pF is off by three orders of

magnitude (560  $\mu$ A through 10 k $\Omega$  is 5.6 V). A 300  $\Omega$ –1 k $\Omega$  pulldown clears it with real margin; the fix is a component VALUE, not a new part.

**What the model does not cover.** The pin-mux gap is real and not papered over: which physical PTxx pins 88/73/07/29/03/05 land on is still a Phase-2 item, so their individual reset pull-state defaults cannot be stated — this model treats every protected gate generically, with no GPIO drive assumed at all except as a deliberately adversarial stimulus. The supervisor IC's own comparator propagation delay (assert side) is not sourced anywhere and is modelled as zero; claim 4's tightest margin (18  $\mu$ s, at the fastest ramp) is the one that delay would eat into first. The inverting stage and kill FET's own gate-drive dynamics are collapsed into one switch with an INFERRED Ron. Ciss=2 nF and Vgs(th)=1.0 V are class figures, not a chosen part's datasheet numbers — no injector/metering FET has been selected yet.

PASS claim 1:  
gate below  
Vgsth, kill  
engaged  
(bound: 10-40  
us)

PASS ...  
actual kill-  
path  
discharge  
time

PASS claim  
1b: gate  
stays below  
Vgsth despite  
an  
adversarial  
driver  
actively  
holding it  
high (GPIO-  
independence)

PASS claim 2:  
final V(gate)  
with the 470  
ohm pulldown  
fitted

PASS ... vs  
the same  
driver with  
no pulldown

PASS ...  
driver output  
impedance  
needed to  
still reach  
9.5 V

PASS ... the  
100 ohm  
driver this  
block assumed  
is too weak  
by

PASS claim 3  
(evidence):  
peak V(gate),  
Crss=20 pF,  
Rpd=10k

PASS claim 3  
(evidence):  
peak V(gate),  
Crss=50 pF,  
Rpd=10k

PASS claim 3  
(evidence):  
peak V(gate),  
Crss=100 pF,  
Rpd=10k

PASS claim 3  
(evidence):  
peak V(gate),  
Crss=200 pF,  
Rpd=10k

PASS claim 3  
(evidence):  
peak V(gate),  
Crss=500 pF,  
Rpd=10k

PASS claim 3  
(evidence):  
peak V(gate),  
Crss=1000 pF,  
Rpd=10k

PASS ... memo  
11's own hand  
arithmetic at ~5.6 mV -- off by 1000x ( $560 \text{ uA} * 10 \text{ kOhm} = 5.6 \text{ V}$ , not 5.6 mV); thi  
Crss=200 pF  
claimed

PASS ... so  
the 10k  
pulldown  
fails even at  
the memo's  
OWN low end

PASS claim 4:  
gate safe  
before 2.97V  
guarantee  
floor,  
ramp=293 us

PASS claim 4:  
gate safe  
before 2.97V  
guarantee  
floor,  
ramp=585 us

PASS claim 4:  
gate safe  
before 2.97V  
guarantee  
floor,  
ramp=1.17 ms

|                                                                                                       |                    |
|-------------------------------------------------------------------------------------------------------|--------------------|
| <div>PASS claim 4:<br/>gate safe<br/>before 2.97V<br/>guarantee<br/>floor,<br/>ramp=10 ms</div>       |                    |
| <div>PASS claim 4:<br/>gate safe<br/>before 2.97V<br/>guarantee<br/>floor,<br/>ramp=100 ms</div>      |                    |
| <div>PASS claim 4:<br/>gate safe<br/>before 2.97V<br/>guarantee<br/>floor,<br/>ramp=1000 ms</div>     |                    |
| <div>PASS ...<br/>tightest<br/>margin is at<br/>the FASTEST<br/>ramp tried</div>                      | 18 us at 293 us -- |
| <div>PASS claim 5:<br/>kill-path<br/>latency vs<br/>one 26.67 ms<br/>cylinder<br/>interval</div>      |                    |
| <div>PASS ...<br/>caveat:<br/>excludes the<br/>watchdog's<br/>own fault-<br/>DETECTION<br/>time</div> | not                |
| <div>PASS claim 6<br/>(sweep): peak<br/>V(gate) at<br/>Crss=500 pF,<br/>Rpd=220 ohm</div>             |                    |
| <div>PASS claim 6<br/>(sweep): peak<br/>V(gate) at<br/>Crss=500 pF,<br/>Rpd=470 ohm</div>             |                    |
| <div>PASS claim 6<br/>(sweep): peak<br/>V(gate) at<br/>Crss=500 pF,<br/>Rpd=1000 ohm</div>            |                    |
|                                                                                                       |                    |

PASS claim 6  
(sweep): peak  
V(gate) at  
Crss=500 pF,  
Rpd=2200 ohm

PASS claim 6  
(sweep): peak  
V(gate) at  
Crss=500 pF,  
Rpd=4700 ohm

PASS ... same  
sweep read at  
Vgsth=2.5 V,  
the  
permissive  
end

PASS ...  
largest  
pulldown that  
holds the  
gate below  
Vgsth

PASS ... cost  
of 470 ohm:  
drive current  
per gate held  
at 10 V

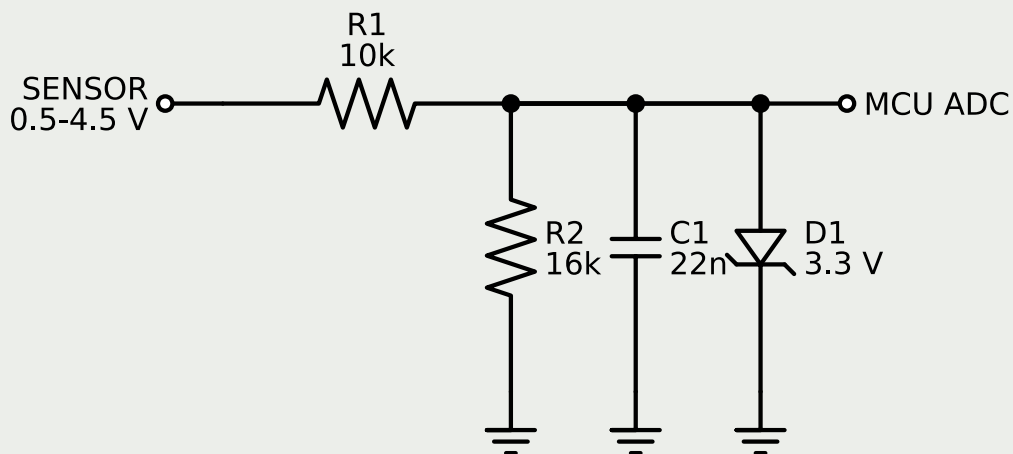
PASS ... and  
across all  
six driver  
gates

Netlist: sim/blocks/supervisor.cir

## 02 **Sensor front-ends**

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Everything that turns something on the engine into a number in the ADC.



The 0.5–4.5 V pressure and position channels, scaled into a 3.3 V ADC.

**Why it looks like that.** Full scale deliberately lands at 2.77 V rather than hard against the ceiling, so the protection clamp can do its job on a harness fault without ever touching the signal in normal running. That costs a little span and buys a front-end 40 V cannot damage. The last four checks cover the ratiometric property itself, which this block was named for and did not model until the spec review caught it: a sensor's output is a fraction of its own supply, so a reading taken against an assumed 5.000 V carries the rail's whole error band — 10% here. Giving the rail its own ADC channel through an identical divider removes it, and the divider ratio cancels algebraically, so its tolerance drops out too.

**What the model does not cover.** Ground offset. Every network here measures against a perfect ground, and pins 34 and 36 are shared returns — which is why spec §4 now specifies a differential front-end for those channels and leaves this one for pin 08, the dedicated return. Also not modelled: divider tolerance matching, on which the rail correction depends, and ADC sampling time against the source impedance.

|                                                                         |                                                                      |
|-------------------------------------------------------------------------|----------------------------------------------------------------------|
| PASS sensor zero<br>(0.5 V) maps to                                     | 0.308 V                                                              |
| PASS sensor full<br>scale (4.5 V)<br>maps to                            | 2.769 V                                                              |
| PASS ... clears<br>the 3.3 V clamp                                      | 0.531 V                                                              |
| PASS span used of<br>ADC range                                          | 74.592 %                                                             |
| PASS low-pass<br>corner                                                 | 1,174.898 Hz                                                         |
| PASS loss at<br>sensor's own 100<br>Hz                                  | -0.031 dB                                                            |
| PASS attenuation<br>at 100 kHz<br>switching                             | -38.596 dB                                                           |
| PASS 40 V harness<br>short clamps pin<br>to                             | 3.308                                                                |
| PASS ... fault<br>current limited<br>to                                 | 3.669 mA                                                             |
| PASS raw reading<br>moves with a<br>+/-5% rail drift                    | 10.000 %                                                             |
| PASS ... so an<br>assumed 5.000 V<br>rail is a 10%<br>error band        | 10.0% of reading, and nothing in the raw value reveals it            |
| PASS corrected<br>reading is flat                                       | 0.000 %                                                              |
| PASS correction<br>recovers the true<br>fraction                        | 0.500                                                                |
| PASS ... divider<br>ratio cancels, so<br>its tolerance<br>drops out too | measurand = $V_{sig}/V_{rail}$ -- k appears in both and divides away |
| Netlist: sim/blocks/sensor_ratiometric.cir                              |                                                                      |

diff amp -- netlist's Bd: gain 16/26 (matches the single-ended ratio) plus a realistic 80 dB CMRR term



Not drawn: Ib (netlist, 20 mA into gs) -- the coolant sensor's own return current, sharing this ground via splice V3. It is what moves gs; Rg is what decides how far it moves.

The same channel measured sensor-to-sensor-ground instead of sensor-to-ECU-ground, swept against the harness return's resistance.

**Why it looks like that.** Pin 34 is spliced between the boost sensor and the coolant sensor, so it carries both sensors' return current. A single-ended input reads the resulting offset as signal. Research memo 09 found this is exactly what Deep Sea Electronics specifies against ( $\pm 2$  V common mode on every sender channel) and what SEDEMAC's dedicated sensor common point exists for. Note what the OEM did with the channel that could not tolerate it: rail pressure gets pin 08, spliced with nothing. The schematic below draws one divider ( $R1/R2$ , the single-ended path) plus a generic amplifier for the differential path,

because that is what the netlist models — a behavioral gain block (Bd: 16/26 gain, matched to R1/R2's own ratio, plus an 80 dB CMRR term), not a second resistor divider. An earlier version of this drawing showed two dividers joined by a plain wire, which matched neither the netlist nor any circuit that would actually reject a ground offset.

**What the model does not cover.** The error is about half a percent of span at 1 ohm of return resistance, and that understates it. It is a bias rather than noise, so it never averages out; it grows as contacts corrode; and it is indistinguishable from a real reading — a boost pressure biased high permits more fuel, and the first symptom is smoke, not a fault code. Rg, the swept harness + connector resistance, is drawn; Ib, the 20 mA coolant-sensor current that actually moves gs, is not — it is a stimulus, not a part of this front end, and it is captioned rather than drawn for that reason.

|                                                      |                                                              |
|------------------------------------------------------|--------------------------------------------------------------|
| PASS clean harness (50 mOhm), single-ended error     | 0.612 mV                                                     |
| PASS 1 ohm return, ground offset present             | 19.903 mV                                                    |
| PASS ... single-ended reading error                  | 12.248 mV                                                    |
| PASS ... differential reading error                  | 0.127 mV                                                     |
| PASS differential is better by                       | 96.448 x                                                     |
| PASS single-ended error at 2 ohm (bad contact)       | 0.995 % span                                                 |
| PASS ... single-ended needs contact resistance below | ~0.1 ohm forever, to hold 0.1% -- differential needs nothing |

Netlist: sim/blocks/sensor\_differential.cir



The thermistor branch of U2, as a pull-up divider and as a constant-current source, across the full  $-40$  to  $150$  C range.

**Why it looks like that.** Memo 09 found DSE drives its temperature senders with a fixed  $15$  mA and measures differentially, and the obvious move is to copy it. Checked against an actual charge-air NTC it does not transfer:  $88$  ohms to  $59$  kilohms is a  $670:1$  span against the  $50:1$  a genset sender covers, and a current source sized to give useful signal at  $150$  C demands  $5.9$  V from a  $5$  V rail at  $-40$  C. The divider's non-linearity is what saves it — it is bounded by the supply at both ends.

**What the model does not cover.** Which branch this engine actually needs is still U2, and one resistance measurement at the machine settles it. The self-heating figure should be revisited once the real sensor's dissipation constant is known.

|                                                |                                                    |
|------------------------------------------------|----------------------------------------------------|
| PASS divider at 150 C (88 ohm)                 | 0.192 V                                            |
| PASS divider at 25 C (2.5k)                    | 2.664 V                                            |
| PASS divider at -40 C (59k)                    | 4.820 V                                            |
| PASS ... never leaves the 5 V rail             | 4.820 V                                            |
| PASS 100 uA source needs this at -40 C         | 5.899 V                                            |
| PASS ... but the rail is 5 V                   | 5.9 V demanded -- out of compliance, reading clips |
| PASS 100 uA source at 150 C gives only         | 8.800 mV                                           |
| PASS current that WOULD stay in compliance     | 84.746 uA                                          |
| PASS ... leaving this at 150 C                 | 7.458 mV                                           |
| PASS divider self-heating power at 25 C        | 2.829 mW                                           |
| PASS ... error at ~2 mW/C dissipation constant | 1.414 degC                                         |
| Netlist: sim/blocks/ntc_frontend.cir           |                                                    |



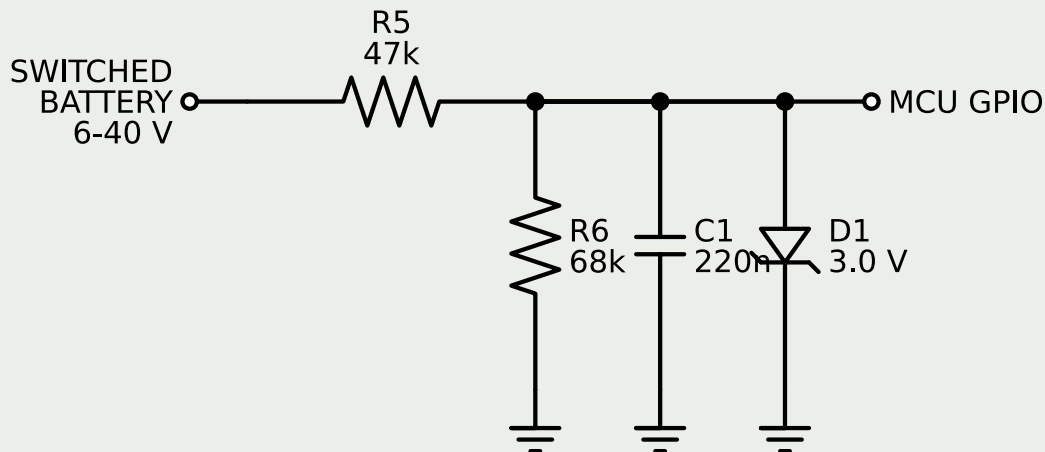
R3/R4: the adopted divider (spec sec.4, pins 04/06).  
battery\_sense.cir also carries a rejected R1/R2 75k/10k  
divider, sized for a 5 V ADC -- not drawn here.

The 6–40 V system rail scaled into the ADC.

**Why it looks like that.** The inherited 75k/10k divider was sized for a 5 V ADC and reaches 4.69 V at 40 V — it would have damaged the input pin. 120k/10k peaks at 3.07 V and still resolves 573 counts at the 6 V cranking dip. The schematic below draws only the adopted 120k/10k divider, and it now carries the netlist's own designators for it, R3/R4 — spec §4 gives the same 120k/10k for pins 04/06. The rejected 75k/10k comparison is R1/R2 in battery\_sense.cir; it is not drawn, since it is not the circuit on the board, and using its designators for the drawn divider (an earlier version of this drawing did) pointed a reader at the wrong resistors.

**What the model does not cover.** Nothing subtle. A resistive divider is exactly linear and the check confirms it, which is there to catch a wrong topology rather than a wrong value.

|                                        |                                      |
|----------------------------------------|--------------------------------------|
| PASS inherited 75k/10k peak at 40 V    | 4.694 V                              |
| PASS ... and so exceeds a 3.3 V ADC    | 4.69 V > 3.3 V -- would damage input |
| PASS corrected 120k/10k peak at 40 V   | 3.069 V                              |
| PASS ... stays under the 3.3 V rail    | 3.069                                |
| PASS headroom below rail               | 0.231 V                              |
| PASS reading at 6 V cranking dip       | 0.462 V                              |
| PASS ... resolves to usable ADC counts | 572.867 LSB                          |
| PASS divider is linear (max residual)  | 0.000 V                              |
| PASS slope matches 10/130              | 0.077                                |
| Netlist: sim/blocks/battery_sense.cir  |                                      |



The switched-battery inputs, active high.

**Why it looks like that.** This one cannot be built as a divider at all. Reading logic high at a 6 V cranking dip needs a ratio above 0.385; staying under 3.3 V at 40 V needs it below 0.0825. No fixed ratio satisfies both, so the original was impossible rather than merely mis-valued. A 3.0 V zener sets the level and the divider only biases into it, which holds the output flat within 92 mV across the whole range.

**What the model does not cover.** The debounce capacitor went from 100 nF to 220 nF after simulation showed the smaller value let two threshold crossings through a bouncing contact — which firmware would have read as two separate presses.

|                                              |          |
|----------------------------------------------|----------|
| PASS never exceeds 3.6 V abs max over 6-40 V | 2.954    |
| PASS logic high at 6 V cranking dip          | 2.863    |
| PASS logic high at 40 V load dump            | 2.954    |
| PASS level is flat across 6-40 V (spread)    | 0.092 V  |
| PASS settled high after bounce ends          | 2.910    |
| PASS threshold crossings during bounce       | 0.000    |
| PASS debounce time constant                  | 6.114 ms |

Netlist: sim/blocks/discrete\_input.cir



healthy 2.86-2.95 V (clamped, flat) -- tripped 0.31-2.07 V (linear, Rt in circuit, never reaches the zener knee) -- wire fault ~0 V (neither Rt nor battery reaches the divider at all)

A three-state supervised loop reading whether the mandatory standalone overspeed trip module's contacts are open or closed — not a copy of the input above, on purpose.

**Why it looks like that.** Spec §3: U12 closed badly on 18 Sep 2026 — the fitted DSE4522's fuel relay only signals, it does not remove power, so a standalone trip module in series with the fuel metering unit's own battery feed (fuse 8F1) is mandatory again, and this ECU must sense its state rather than read a trip as a fuelling anomaly. `discrete_input.cir`'s own accepted limitation — an open wire reads identically to 'not asserted', accepted there because changing the sense would break OEM harness compatibility — does not carry over: this pin has no OEM harness to stay compatible with, and an undetectable broken wire here would read 'not tripped' at the exact moment a trip occurs, which is the one failure spec §3 exists to catch. So this block adds an end-of-line resistor (Rt, 1.2 MΩ) in parallel with the module's own auxiliary contact, physically at the module end of the new harness run, giving three distinguishable ADC bands on the same R5/R6/zener front end `discrete_input.cir` already validates: healthy (contact closed, Rt shorted) clamps to 2.86–2.95 V exactly as that file's input does; tripped (contact open, Rt in circuit) is a plain, unclamped linear divider that stays 0.31–2.07 V across 6–40 V, never reaching the zener's knee; wire fault (a break anywhere in the run removes both the contact and Rt the same way) reads ~0 V. A contact-bounce transient, reusing `discrete_input.cir`'s own nasty bounce profile inverted, shows the reading never leaves the healthy band during the bounce itself — C1's

time constant on this branch is long enough to filter chatter shorter than a few hundred microseconds — and settles cleanly into the tripped band only once the contact stays open.

**What the model does not cover.** No trip module is chosen (memo 02's costed GAC SSW675 / Murphy HD9063 candidates survive costing but neither's aux-contact rating is pulled), so  $R_t=1.2\text{ M}\Omega$  is this file's own INFERRED choice with no datasheet or industry EOL-loop convention behind it. The loop is high impedance (the same few-hundred-microamp class `discrete_input.cir`'s divider runs at), not the low-Z, milliamp-class loop industrial supervised-contact practice normally uses — chosen to reuse the existing front-end rather than build a new one, and the cost is a genuinely tight margin: at the 6 V cranking dip with  $R_t$  at +5% tolerance, the tripped band's floor sits only 0.147 V above the wire-fault threshold, about half the nominal-case margin, before any harness leakage is stacked on top. This pin also needs to land on an ADC-capable channel, not a bare GPIO — a pin-mux consideration for whoever assigns the physical PTxx map.

PASS

healthy  
(contact  
closed)  
band floor

PASS

healthy  
band  
ceiling

PASS

tripped  
(Rt in  
circuit)  
band  
floor, at  
6V  
cranking  
dip

PASS

tripped  
band  
ceiling,  
at 40V

PASS wire  
fault  
reads near  
zero

PASS

tripped  
band never  
reaches  
the  
zener's  
3.0V knee

PASS

margin:  
wire-fault  
threshold  
to tripped  
floor  
(nominal  
Rt)

PASS

margin:  
tripped  
ceiling to  
healthy  
floor  
(nominal  
Rt)

PASS ...  
both bands  
separated  
with real  
margin,  
not  
touching

wire-fault margin

PASS  
during  
bounce  
( $t < 4.5\text{ms}$ ),  
reading  
never  
leaves the  
healthy  
band

PASS  
settled  
value  
( $t = 100\text{ms}$ )  
approaches  
the DC  
asymptote  
(hand:  
 $0.6206\text{ V}$ )

PASS ...  
which sits  
inside the  
tripped  
band, not  
wire-fault  
or healthy

PASS ...  
global  
minimum  
over the  
whole run  
stays  
clear of  
wire-fault

PASS  
open-wire  
detection:  
needed  
here,  
unlike  
pins  
20/24/71

no OEM harness to stay compatible with, and an undetectable open reads

Netlist: sim/blocks/trip\_module\_sense.cir

## 03 **Speed and position**

---

The most safety-critical input on the board.

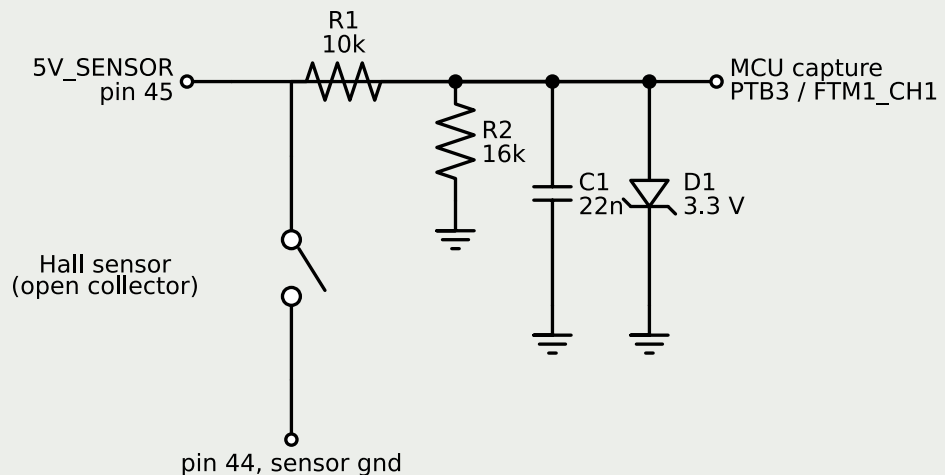


A variable-reluctance sensor into a zero-crossing comparator, run at both running speed and cranking speed.

**Why it looks like that.** A VR sensor generates voltage by induction, so its amplitude is proportional to speed: roughly 20 V peak at 1500 rpm and 2 V at cranking. A fixed 5 V threshold, which looks perfectly sensible against a 20 V signal, produces all 30 teeth at running speed and NOTHING at cranking. That is the worst failure shape there is — it works on the bench and the engine never starts. Zero-crossing detection does not care about amplitude, and the crossing is also where the waveform is steepest and timing jitter is smallest.

**What the model does not cover.** A real VR signal is not a clean sine — tooth edges distort it and the missing-tooth sync gap produces one large excursion per revolution. Amplitude at true cranking speed on a cold engine with a weak battery can be under a volt, which is when starting matters most, and the hysteresis window needs checking against the real sensor rather than this 2 V estimate.

|                                                                                                                        |              |
|------------------------------------------------------------------------------------------------------------------------|--------------|
| PASS zero-cross,<br>running (1500 rpm,<br>30 teeth)                                                                    | 30.000 edges |
| PASS zero-cross,<br>cranking (150 rpm,<br>3 teeth)                                                                     | 3.000 edges  |
| PASS fixed 5 V<br>threshold, running                                                                                   | 30.000 edges |
| PASS fixed 5 V<br>threshold, cranking                                                                                  | 0.000 edges  |
| PASS ... so a fixed<br>threshold never works at 1500 rpm, blind at 150 rpm -- passes on the bench<br>starts the engine |              |
| PASS running tooth<br>period                                                                                           | 666.667 us   |
| PASS ... jitter<br>across the window                                                                                   | 0.000 us     |
| Netlist: sim/blocks/vr_conditioner.cir                                                                                 |              |



R1/R2/C1/D1: identical values to sensor\_ratiometric.cir's own 5V\_SENSOR front end -- same divider ratio, same filter corner, same clamp. Not a coincidence: pin 46 is not 5 V tolerant on this MCU (S32K1xx datasheet  $V_{ih\ max} = V_{DD} + 0.3\ V$ ), so the spec's literal 'pull-up to 5V\_SENSOR' has to be divided into the 3.3 V domain before PTB3, the same treatment every other 5 V-rail input gets.

A cam Hall sensor's open-collector output, pulled up and divided into the 3.3 V domain before PTB3/FTM1\_CH1, instead of the spec's literal pull-up straight to the 5 V sensor rail.

**Why it looks like that.** PTB3 is not 5 V tolerant for anything this design should rely on — the S32K148's own datasheet puts guaranteed valid-high recognition at  $V_{DD} + 0.3\ V$ , about 3.6 V, and treats a higher DC injection allowance as a fault margin, not a design basis for a pin that sees 5 V every cam revolution for the machine's whole life. Nothing in sim/blocks/ ever modelled this pin before — there is a conditioner for the crank sensor and no cam equivalent — so the overvoltage sat uncaught. The divider (10k/16k) and filter (22 nF, 1176 Hz corner) reuse sensor\_ratiometric.cir's own values exactly, since both share the same 5V\_SENSOR rail and the same target domain: 94x above the cam's 12.5 Hz signal (half crank speed at 1500 rpm, one lobe per revolution assumed) and still well below switching-frequency noise.

**What the model does not cover.** The real Hall sensor's saturation voltage, sink current, and internal pull-up (if any) are not modelled — no part chosen. Lobe count is INFERRED at 1 per cam revolution; the TOLERANCE section shows the filter margin survives a 3-lobe alternative just as well. A real cam signal's edge shape and any duty-cycle asymmetry from target

geometry are not modelled, the same limitation  
vr\_conditioner.cir already states for the crank signal's own non-ideal waveform.

|                                                                       |             |
|-----------------------------------------------------------------------|-------------|
| PASS sensor off, 5V_SENSOR nominal (5.0 V) → pad                      | 3.072 V     |
| PASS ... clears the 3.6 V abs max                                     | 0.528 V     |
| PASS sensor_rail.cir's own fault floor (4.5 V) → pad                  | 2.769 V     |
| PASS ... still reads logic high ( $\geq 2.31$ V, $0.7 \cdot V_{DD}$ ) | 2.769 V     |
| PASS above-nominal rail (5.25 V) → pad                                | 3.147 V     |
| PASS ... still clears the 3.6 V abs max                               | 0.453 V     |
| PASS never exceeds 3.6 V abs max in transient                         | 3.072 V     |
| PASS rising edges, 170 ms window at 12.5 Hz (80 ms period)            | 2.000 edges |
| PASS 40 V harness short on signal line clamps pad to                  | 3.308 V     |
| PASS ... fault current limited to                                     | 3.669 mA    |

Netlist: sim/blocks/cam\_frontend.cir

## 04 Injection

---

The most expensive stage on the board and the reason it exists.



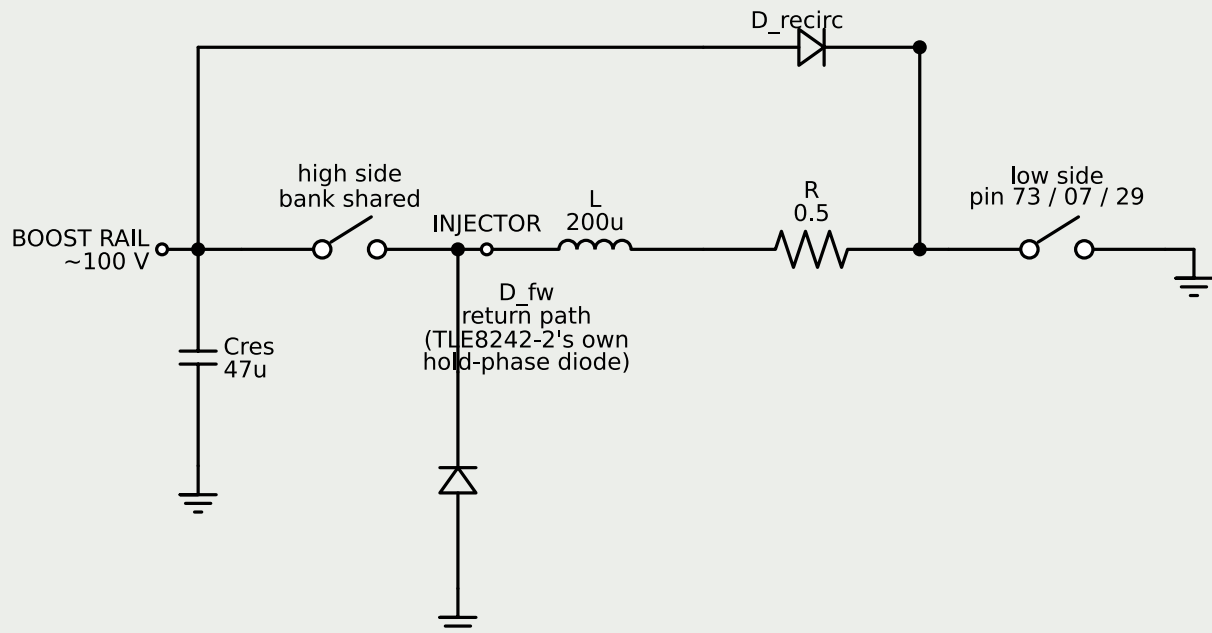
The same injector driven from the battery and from a 100 V rail, compared on time to reach the 18 A peak threshold.

**Why it looks like that.** 439 us from the battery, 38 us from the boost rail. On a 1 ms injection the battery drive spends 44% of the event just lifting the needle; the boost rail spends 3.8%. That ratio is the entire argument for the stage.

**What the model does not cover.** An RL model of the injector with no magnetic saturation and no moving needle. The envelope it sits in — 65–115 V, 12–24 A peak — is family-wide from memo 04, not this injector's measured values. That is U5. Also not modelled: the high/low-side switches and the sense resistor, dashed in the schematic — this is a turn-on-only step response straight onto L/R, and R already stands in for whatever series resistance the real drive path adds.

|                                                       |            |
|-------------------------------------------------------|------------|
| PASS time to 18 A peak from 13.5 V battery            | 439.496 us |
| PASS time to 18 A peak from 100 V boost               | 37.796 us  |
| PASS boost is faster by                               | 11.628 x   |
| PASS battery drive eats this much of a 1 ms injection | 43.950 %   |
| PASS boost drive eats this much of a 1 ms injection   | 3.780 %    |
| FAIL ... boost is faster by, -40C ambient             | 9.856 x    |
| PASS ... time to 18 A from 100 V boost, -40C ambient  | 37.296 us  |
| FAIL ... boost is faster by, +125C ambient            | 19.017 x   |
| PASS ... time to 18 A from 100 V boost, +125C ambient | 38.496 us  |

Netlist: sim/blocks/injector\_boost.cir



18 A  $\rightarrow$   $\sim$ 0 A in  $\sim$ 33  $\mu$ s, rail bump  $\sim$ +6.4 V/event (measured)  
 memo 10 predicted  $\sim$ 36  $\mu$ s /  $\sim$ +6.9 V ideal -- real diode and switch losses shave both, and the bump undershoots the 7.3 V on-phase droop it is paired with -- net loss, not gain

The low-side switch opening at 18 A, with a recirculation diode routing the coil's current back into the same boost cap (Cres, 47  $\mu$ F) instead of a clamp that just burns it — plus a second diode (D\_fw) the loop needs to close, and a comparison against freewheeling to the battery instead.

**Why it looks like that.** Research memo 10 surveyed real injector-driver topologies and ruled out a Zener/active clamp (Nexperia AN50003 flags repetitive active clamp as reliability-“Questionable”) and a bare battery freewheel (too slow) in favor of recirculation to the existing boost rail — confirmed production diesel-injector practice via a 1988 Marelli Autronica patent, not a textbook flourish. Simulated: 33  $\mu$ s to collapse 18 A to zero (memo predicted  $\sim$ 36  $\mu$ s), 25x faster than freewheeling to the battery (830  $\mu$ s — slower than the memo's own simplified estimate, because a real RL circuit's exponential tail matters more at 13.5 V than at 100 V). The rail bump per event is +6.4 V, under the memo's +6.9 V ideal because real diode and switch losses are modelled here, and it undershoots the 7.3 V on-phase droop it is paired with — net loss, not gain, so three back-to-back events never push the rail over its 100 V setpoint. But that verdict hinges on which current is being cut off: boost\_converter.cir's own arrangement A has the battery supply HOLD current, not the boost rail, and recirculating a hold-current cutoff through the same diode has no offsetting boost-side draw to net against

— modelled separately here, it overshoots to 102 V on a single event. boost\_converter.cir's charge regulation (Bchg) has no bleed path either way; it just happens not to need one for the peak-only case.

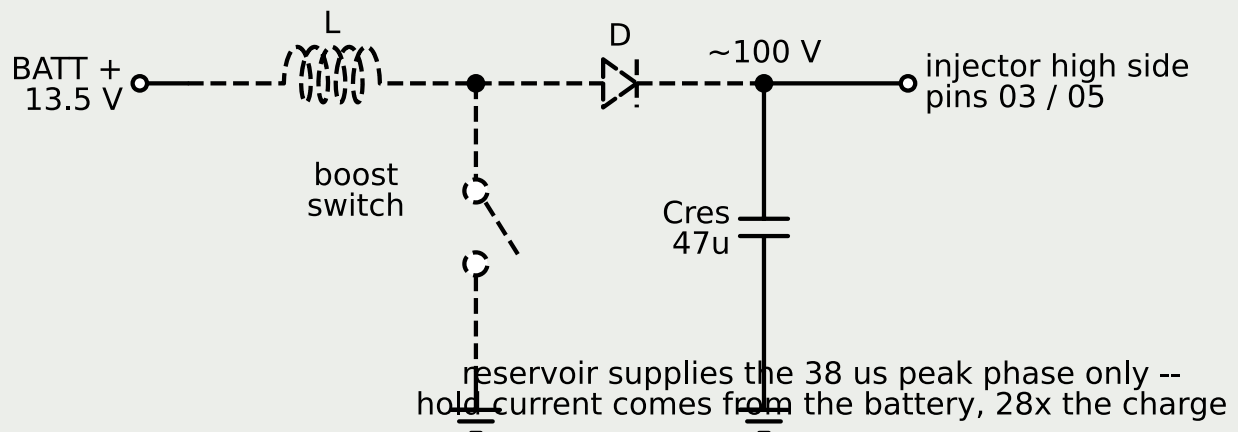
**What the model does not cover.** The differential-mode EMI edge (memo's own claim 5) is not modelled — the switches here are ideal with zero transition time, so any  $dV/dt$  read off this simulation would be a solver artifact, not a FET's real slew rate. Also not modelled: diode reverse recovery ( $Q_{rr}$ ), the diode's thermal transient, and — inherited from injector\_boost.cir — magnetic saturation and needle motion.

|                                                                                                                   |                                                             |
|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| PASS turn-off time,<br>18 A → ~0 A on the<br>boost rail                                                           |                                                             |
| PASS ... vs<br>freewheel-to-battery<br>baseline (comparison<br>only)                                              |                                                             |
| PASS ... boost<br>recirculation is<br>faster by                                                                   |                                                             |
| PASS on-phase droop,<br>pilot event                                                                               |                                                             |
| PASS ... matches<br>boost_converter.cir's<br>own droop figure                                                     |                                                             |
| PASS recovery bump,<br>pilot event<br>(isolated)                                                                  |                                                             |
| PASS ... undershoots<br>the droop it is<br>paired with (real<br>losses)                                           |                                                             |
| PASS rail never<br>exceeds 100 V<br>setpoint (3-event<br>circuit)                                                 |                                                             |
| PASS rail recovered<br>by the next cylinder<br>(26.67 ms after post)                                              |                                                             |
| PASS hold-current<br>cutoff into the SAME<br>cap, no offsetting<br>draw                                           |                                                             |
| PASS ... THIS is the<br>scenario that<br>overshoots 100 V                                                         |                                                             |
| PASS ... so the<br>verdict hinges on<br>whether hold-current<br>cutoffs share the<br>boost-recirculation<br>diode | peak-only recirculation never overshoots; hold-current reci |
| PASS peak<br>instantaneous power<br>in D_recirc during<br>turn-off                                                |                                                             |
| PASS ... exceeds the<br>~3.6 W worst-case<br>AVERAGE by                                                           |                                                             |
|                                                                                                                   |                                                             |

PASS differential-  
mode edge vs  
emi\_filter.cir

not modelled here (ideal switches) --

Netlist: sim/blocks/injector\_turnoff.cir



The reservoir capacitor under a real injection event, with and without the hold phase drawn from the same rail.

**Why it looks like that.** The converter cannot deliver 18 A — it delivers a trickle and the capacitor delivers the pulse, so the capacitor is the component that matters. The design question is where the HOLD current comes from, and the answer is not subtle: hold costs 29 times the peak phase's charge, so a rail supplying both collapses completely. Hold comes from the battery. 50 mA of average charging recovers the rail with room to spare in the 26.67 ms before the next cylinder fires.

**What the model does not cover.** Charge balance only. L, D and the switch are dashed in the schematic because the converter physically has them, but there is no inductor, no switch and no control loop in the netlist — just a current source standing in for the converter's average output. It says nothing about the converter's stability or its inrush, and the collapsed voltage in the second case should be read as 'far below usable' rather than as a number.

|                                                 |                                           |
|-------------------------------------------------|-------------------------------------------|
| PASS droop, peak phase only (47 uF)             | 7.255 V                                   |
| PASS ... rail stays usefully above battery      | 92.745 V                                  |
| PASS droop, peak AND hold from the rail         | 210.913 V                                 |
| PASS ... so the rail collapses                  | -111 V -- hold must come from the battery |
| PASS hold phase costs this much more charge     | 29.071 x                                  |
| PASS recovered by the next injection (26.67 ms) | 100.000 V                                 |
| PASS ... 50 mA average charging is enough       | 100.000 V                                 |
| Netlist: sim/blocks/boost_converter.cir         |                                           |

## 05 **Actuators**

---

Outputs that move something.



Low-side PWM into the metering solenoid at three frequencies, all at the same duty.

**Why it looks like that.** Rail pressure is controlled by the AVERAGE current through this coil, and all three frequencies produce the same average — which is what makes the control law independent of the PWM constant. What changes is ripple, and ripple is not purely a nuisance here: a proportional solenoid has stiction, and deliberate dither keeps it moving. At 100 Hz the period is comparable to the coil's own 3 ms time constant and the result is 152% ripple — that is not dither, that is chopping, and the valve follows the PWM rather than its average.

**What the model does not cover.** The correct dither amplitude is a property of the specific metering unit's mechanics and is manufacturer data tied to a part number — the same gap as U5, in a smaller form. Keeping the frequency a firmware constant

rather than a hardware one means it can be tuned on the machine. Coil inductance is also not constant: it falls as the armature moves and as the core saturates.

|                                                      |                                      |
|------------------------------------------------------|--------------------------------------|
| PASS mean current at 100 Hz                          | 0.639 A                              |
| PASS mean current at 1 kHz                           | 0.639 A                              |
| PASS mean current at 10 kHz                          | 0.634 A                              |
| PASS ripple at 100 Hz                                | 0.969 A                              |
| PASS ... as fraction of mean                         | 151.786 %                            |
| PASS ripple at 1 kHz                                 | 0.118 A                              |
| PASS ripple at 10 kHz                                | 0.012 A                              |
| PASS ripple scales as 1/f                            | 9.974 x                              |
| PASS 100 Hz is not dither, it is chopping            | 152% ripple -- valve follows the PWM |
| FAIL ... mean current at 100 Hz, -40C ambient        | 0.862 A                              |
| FAIL ... mean current at 1 kHz, -40C ambient         | 0.863 A                              |
| FAIL ... mean current at 10 kHz, -40C ambient        | 0.858 A                              |
| PASS ... control law: frequencies still agree, -40C  | 0.514 %                              |
| FAIL ... mean current at 100 Hz, +125C ambient       | 0.463 A                              |
| FAIL ... mean current at 1 kHz, +125C ambient        | 0.463 A                              |
| FAIL ... mean current at 10 kHz, +125C ambient       | 0.459 A                              |
| PASS ... control law: frequencies still agree, +125C | 0.925 %                              |
| Netlist: sim/blocks/metering_unit_pwm.cir            |                                      |



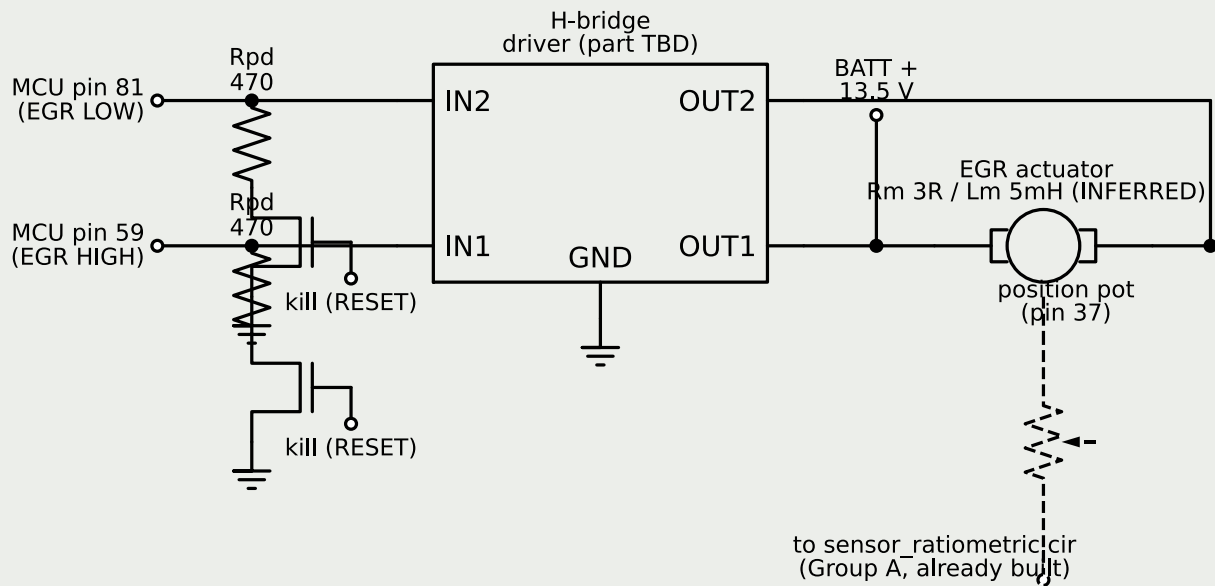
A low-side FET switching a relay coil, run with and without the flyback diode.

**Why it looks like that.** With the diode the drain stays at 14.3 V on turn-off. Without it the coil's stored energy has nowhere to go except through the FET.

**What the model does not cover.** The simulated peak without the diode is hundreds of kilovolts, which is the ideal-switch model taken literally. A real MOSFET avalanches at its breakdown voltage and dissipates that energy in the die instead — once per relay operation, until it fails. The magnitude is a model artifact; the conclusion is not.

|                                       |               |
|---------------------------------------|---------------|
| PASS drain peak WITH flyback diode    | 14.274 V      |
| PASS drain peak WITHOUT diode         | 402,899.522 V |
| PASS ... diode suppresses the kick by | 28,226.510 x  |

Netlist: sim/blocks/relay\_driver.cir



dashed: on the board, not in this netlist -- the driver's internal FETs are modelled generically (four switches + body diodes), the position pot is a REAL front end but it is sensor\_ratiometric.cir's, not a new one (see that block's pins 41/35/80/37)

Two logic/PWM lines into an external H-bridge driving a positional DC-motor actuator both ways — the one output on this board that is not a low-side switch — modelled as four generic switch positions with anti-parallel body diodes, plus the same active kill-clamp (Rpd 470  $\Omega$ , keyed to the supervisor's RESET) spec §4 requires at every driver gate.

**Why it looks like that.** This is a positional actuator, not an on/off solenoid: it is held at a commanded position, which is why it needs a bridge. Spec §4's outputs table lists pins 59/81 in the same row set as the six gates supervisor.cir already protects, but that file's own header enumerates only 88/73/07/29/03/05 — EGR was never added because this block did not exist. Reusing supervisor supervisor.cir's validated kill topology closes that gap rather than re-deriving it. The two-input "sign" scheme this design assumes (IN1 drives the S1+S4 diagonal, IN2 drives S3+S2) means no combination the ECU can present ever commands one leg's own high and low side together — real shoot-through is a property of the eventual driver IC's internal interlock, a part not yet chosen, not something these two lines can cause on their own. Stall current (jammed, or simply sitting at its own end of travel — normal operation for a positional actuator, not a fault) comes out to 4.36 A at an assumed 3  $\Omega$  armature resistance, comfortably past a class-typical 2–3 A small-

actuator driver rating: the eventual part needs active current limiting, not a fixed series resistor. Commanding coast with current already flowing is checked with and without the body diodes, the same comparison relay\_driver.cir runs for its flyback diode: with them the bridge nodes stay within a diode drop of the rail; without them the ideal-switch model produces a multi-megavolt artifact, because the motor's stored energy has nowhere else to go.

**What the model does not cover.** No H-bridge driver IC or FET is chosen —  $R_{on}$ , the body-diode figures, and the motor's  $R_m/L_m$  are all INFERRED class figures, the same gap as every other unselected FET on this board. Whether the eventual driver is integrated or discrete FETs behind a gate driver is undecided (memo 12 settled this for the injector and metering stages but not for EGR), so the  $470\ \Omega$  kill-clamp value here is a uniform-standard choice, not a fresh Miller-coupling derivation for the 13.5 V domain EGR actually runs in. No mechanical model of the actuator exists — no return-spring constant, no gear ratio, no end-stop switches — so "coast lets the spring close the valve" is stated as the assumed fail-safe direction, not simulated. Position feedback on pin 37 is not modelled in this file at all: it is Group A ratiometric, already built and checked in sensor\_ratiometric.cir, and building a second front end here would duplicate a proven circuit rather than reuse it.

|                                                                                                        |                                                         |
|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| PASS kill-clamp: gate<br>below Vgsth (hand<br>estimate 22.8 ns)                                        | 22.                                                     |
| PASS ... adversarial<br>driver (100R/10V)<br>still held below Vgsth                                    | 0                                                       |
| PASS ... adversarial<br>settle voltage (hand:<br>$10 \times 4.95 / 104.95 = 0.471$<br>V)               | 0                                                       |
| PASS stall current<br>(Vemf=0, hand:<br>13.5/3.1)                                                      | 4                                                       |
| PASS running current<br>(Vemf=9V, hand:<br>4.5/3.1)                                                    | 1                                                       |
| PASS coast current<br>(IN1=IN2=0)                                                                      | 0                                                       |
| PASS stall current<br>exceeds a class-<br>typical 2-3 A driver<br>rating                               | 4                                                       |
| PASS ... so the<br>eventual driver needs<br>active current<br>limiting, not a fixed<br>series resistor | 4.35 A at Rm=3 ohm alone exceeds a 2-3 A class          |
| PASS forward steady<br>current at t=19.9ms<br>matches section 2's<br>stall figure                      | 4                                                       |
| PASS WITH diodes: low<br>side stays within a<br>diode drop of ground<br>(hand: -1.09 V)                | -1                                                      |
| PASS WITH diodes:<br>high side stays within<br>a diode drop of the<br>rail (hand: 14.59 V)             | 14                                                      |
| PASS WITHOUT diodes:<br>low side swings far<br>past ground                                             | 9,038,255                                               |
| PASS WITHOUT diodes:<br>high side swings far<br>past the rail                                          | 9,038,269                                               |
| PASS ...<br>recirculation diodes<br>are load-bearing, not<br>decorative                                | bounded to 14.6 V with diodes vs $9.04 \times 10^6$ V w |

|                                                                                                                                                                                       |                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <div>PASS position<br/>feedback (pin 37) is<br/>Group A ratiometric,        0.308-2.769 V into the 3.3 V ADC, see check_sensor_ratiome<br/>proven in<br/>sensor_ratiometric.cir</div> |                               |
| <div>PASS rejected naive<br/>wiring: rail current<br/>with IN1=IN2 asserted</div>                                                                                                     | 270                           |
| <div>PASS ... matching the<br/>two-legs-of-2x50m0hm<br/>hand figure</div>                                                                                                             | 270                           |
| <div>PASS ADOPTED: decoded<br/>DIR/PWM driver, both<br/>inputs asserted</div>                                                                                                         | 4                             |
| <div>PASS ... so decode<br/>removes the fault<br/>state entirely</div>                                                                                                                | 270 A naive vs 4.35 A decoded |
| Netlist: sim/blocks/egr_hbridge.cir                                                                                                                                                   |                               |

## 06 Communications

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The bus to the genset controller.



A split 60+60 termination with a capacitor to ground at the midpoint, against a single 120 ohm resistor.

**Why it looks like that.** Both are 120 ohms differential and a schematic label cannot tell them apart. The difference is common mode: a single resistor gives noise on both wires together nowhere to go, while the split version shunts it to ground without touching the differential signal, because the midpoint is a virtual ground for differential drive. That matters more here than on a truck. The fitted controller is a Deep Sea DSE4522, and DSE's manual is explicit that with a CAN engine the ECU transmits engine speed to it — so this bus carries a safety-relevant value, and it runs beside 415 V three-phase wiring and contactor coils in the same panel.

**What the model does not cover.** Termination is not the whole common-mode story; the choke is the other half and needs a layout. And this does not decide whether to fit termination at all: a 120 ohm terminator belongs at each END of the backbone, and if this ECU is not an end node, fitting one makes the bus worse. That is U4 — so the board should carry the parts with a link and let the machine decide. The DSE4522 terminates CAN on its terminals 18/19/20 and DSE specify 120 ohm cable, which settles the cable but not which nodes carry the terminators.

|                                                                     |              |
|---------------------------------------------------------------------|--------------|
| PASS split termination, differential at 250 kbit/s                  | 119.993 ohm  |
| PASS single 120R, differential at 250 kbit/s                        | 119.993 ohm  |
| PASS ... identical differentially                                   | 0.000 ohm    |
| PASS split termination, common-mode at 10 MHz                       | 30.189 ohm   |
| PASS single 120R, common-mode at 10 MHz                             | 500.000 kohm |
| PASS ... split shunts common mode better by                         | 16,562.490 x |
| PASS split common-mode at the bit rate stays high                   | 138.099 ohm  |
| FAIL ... at 200 ppm/C resistor tempco (matched direction, -40/125C) | 123.952 ohm  |
| Netlist: sim/blocks/can_termination.cir                             |              |